Determining linear equations

What to do

1 Calculate the gradient of the straight line between the following pairs of points.

a (0, 3) and (3, 6) b (1, 5) and (4, 2) c (−5, −3) and (1, −1)

d (6, 2) and (−3, 0) e (−7, 5) and (−5, 7) f (−8, 3) and (−4, 1)

2 Write the equation for the straight line that has:

a a gradient of 2 and passes through the point (0, −7)

b a gradient of −3 and passes through the point (0, 5)

c a gradient of −1 and passes through the point (3, 0).

3 Consider the following equations.

a $y = -2x + 3$ b $y + 3x = 4$ c $3y = 2x + 9$

i Write the equation in the form of $y = mx + c$.

ii State the gradient of the line.

iii State the y-intercept of the line.

4 Answer the following questions for each of these pairs of points.

a (0, 0) and (2, 2)

b (1, 4) and (−1, 6)

i On a set of Cartesian axes, plot the points and draw a line through them.

ii Find the gradient of the line.

iii Find the y-intercept of the line.

iv Use your answers from **ii** and **iii** to write an equation for the line.

v Find one more point that lies on the line.

5 a i Graph a line with a gradient of 4 and a y-intercept of 3.

ii Identify two points on your line.

iii Write an equation for the line

b i On the same set of axes graph another line with a gradient of 4 passing through (0, −2).

ii Write an equation for the line.

c What is the name given to these two lines?

6 Descriptions of three possible lines are given below.

a A line that does not pass through the first quadrant

b A line that passes through exactly two quadrants

c A line which passes through only one quadrant

i Decide whether for each of the above such a line exists. Explain.

ii If a line exists, write a possible equation of the line.

7 Does the point (3, −1) lie on the line $y = -2x + 5$?